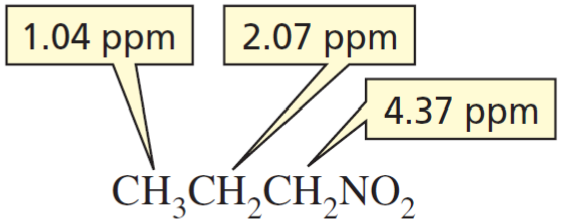
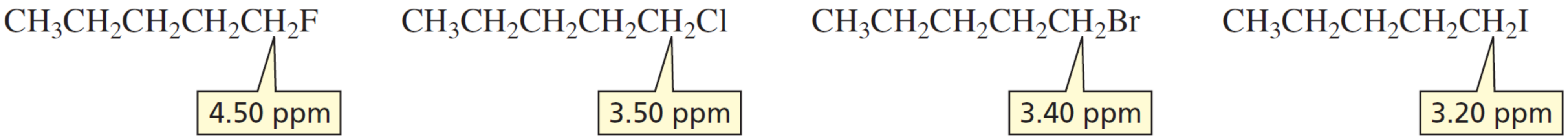
NMR Spectroscopy

THE RELATIVE POSITIONS OF ^{1}H NMR SIGNALS



We expect the ^{1}H NMR spectrum of **1-nitropropane** to have **three signals** because the compound has three different kinds of protons. The closer the protons are to the **electron withdrawing nitro group**, **the less** they are **shielded** from the applied magnetic field, so the higher the frequency at which their signal will appear. Thus, the **protons closest to the nitro group** show a signal at the **highest frequency** (4.37 ppm), and **the ones farthest from the nitro group** show a signal at the **lowest frequency** (1.04 ppm).

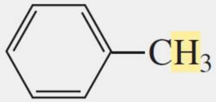
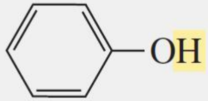
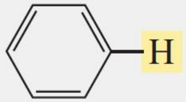
Compare the ***chemical shifts*** of the **methylene protons** immediately adjacent to the halogen in the following alkyl halides.
The position of the signal depends on the **electronegativity of the halogen**—as the electronegativity of the halogen increases, the shielding of the protons decreases, so the frequency of the signal increases. Thus, the signal for the **methylene protons adjacent to fluorine** (the most electronegative of the halogens) occurs at the **highest frequency**, whereas the signal for the methylene protons adjacent to iodine (the least electronegative of the halogens) occurs at the lowest frequency.



NMR Spectroscopy

VALUES OF CHEMICAL SHIFTS

Table 15.1 Approximate Values of Chemical Shifts for ^{1}H NMR[a]

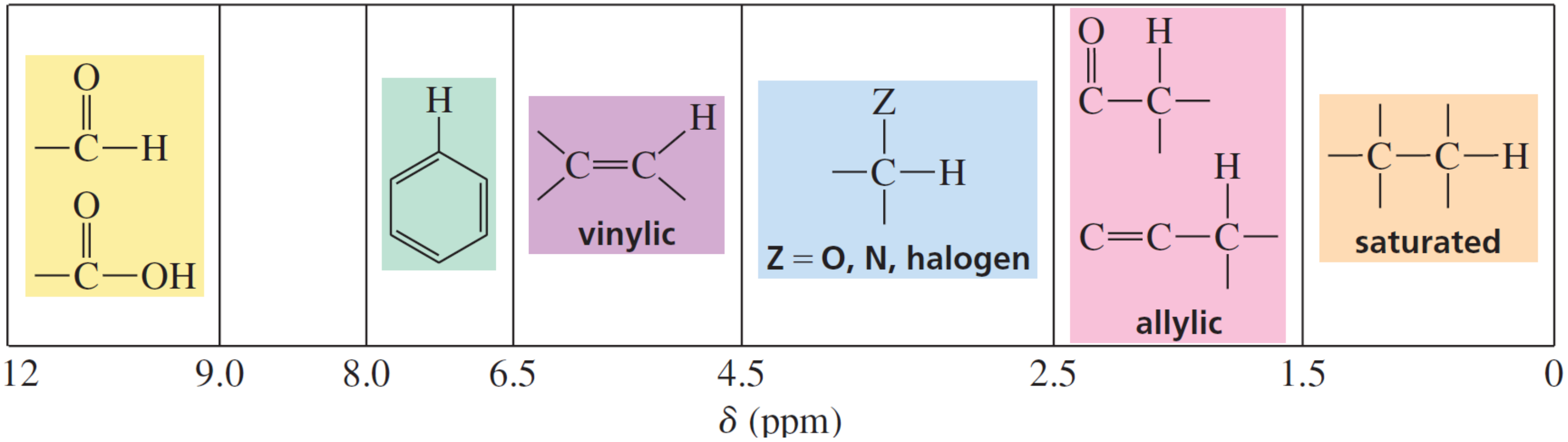
Type of proton	Approximate chemical shift (ppm)	Type of proton	Approximate chemical shift (ppm)
$-\mathrm{CH_3}$	0.85	$\mathrm{I-\overset{\vert}{\underset{\vert}{C}}-H}$	2.5–4
$-\mathrm{CH_2}-$	1.20	$\mathrm{Br-\overset{\vert}{\underset{\vert}{C}}-H}$	2.5–4
$-\overset{\vert}{\mathrm{C}}\mathrm{H}-$	1.55	$\mathrm{Cl-\overset{\vert}{\underset{\vert}{C}}-H}$	3–4
$-\underset{\vert}{\mathrm{C}}=\underset{\vert}{\mathrm{C}}-\mathrm{CH_3}$	1.7	$\mathrm{F-\overset{\vert}{\underset{\vert}{C}}-H}$	4–4.5
$-\overset{\mathrm{O}}{\overset{\Vert}{\mathrm{C}}}-\mathrm{CH_3}$	2.1	$\mathrm{R-NH_2}$	Variable, 1.5–4
	2.3	$\mathrm{R-OH}$	Variable, 2–5
$-\mathrm{C}\equiv\mathrm{C}-\mathrm{H}$	2.4		Variable, 4–7
			6.5–8
$\mathrm{R-O-CH_3}$	3.3	$-\overset{\mathrm{O}}{\overset{\Vert}{\mathrm{C}}}-\mathrm{H}$	9.0–10
$\mathrm{R-\underset{\underset{R}{\vert}}{C}=CH_2}$	4.7	$-\overset{\mathrm{O}}{\overset{\Vert}{\mathrm{C}}}-\mathrm{OH}$	Variable, 10–12
$\mathrm{R-\underset{\underset{R}{\vert}}{C}=\underset{\underset{R}{\vert}}{C}-H}$	5.3	$-\overset{\mathrm{O}}{\overset{\Vert}{\mathrm{C}}}-\mathrm{NH_2}$	Variable, 5–8

[a]The values are approximate because they are affected by neighboring substituents.

NMR Spectroscopy

THE CHARACTERISTIC VALUES OF CHEMICAL SHIFTS

An ^{1}H NMR spectrum can be divided into seven regions, one of which is empty. If you can remember the kinds of protons that appear in each region, you will be able to tell what kinds of protons a molecule has from a quick look at its NMR spectrum

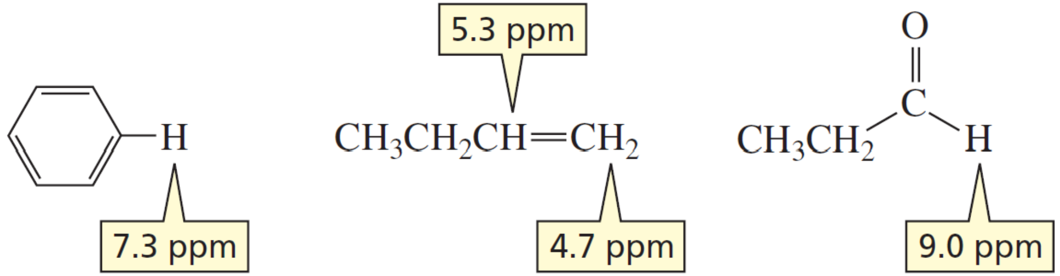


NMR Spectroscopy

DIAMAGNETIC ANISOTROPY

The chemical shifts of hydrogens bonded to ***sp*²** **carbons are at higher** frequencies than one would predict from the electronegativities of the *sp*² carbons.

For example, a hydrogen bonded to a **benzene ring appears at 6.5 to 8.0 ppm**, a hydrogen bonded to the terminal *sp*² carbon of an **alkene appears at 4.7 to 5.3 ppm**, and a hydrogen bonded to a carbonyl carbon appears at **9.0 to 10.0 ppm**

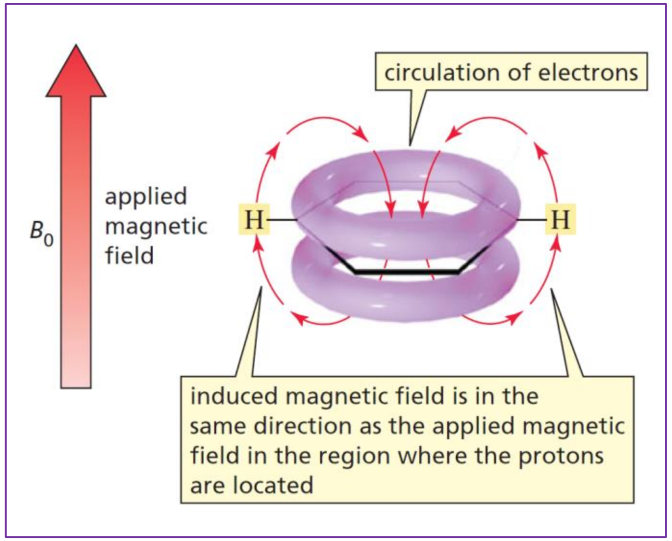


The unusual chemical shifts associated with hydrogens bonded to carbons that form π bonds are due to **diamagnetic anisotropy .** This term describes an environment in which **different magnetic fields** are found at **different points in space**. (*Anisotropic* is Greek for “different in different directions.”)

NMR Spectroscopy

DIAMAGNETIC ANISOTROPY

Because π electrons are less tightly held by nuclei than are **σ** electrons, π **electrons are freer to move** in response to a magnetic field. When a magnetic field is applied to a compound with π electrons, the π electrons move in a circular path that induces a **small local magnetic field**. How this induced magnetic field affects the chemical shift of a proton depends on the **direction of the induced magnetic field in the region** where the proton is located, relative to the direction of the applied magnetic field.

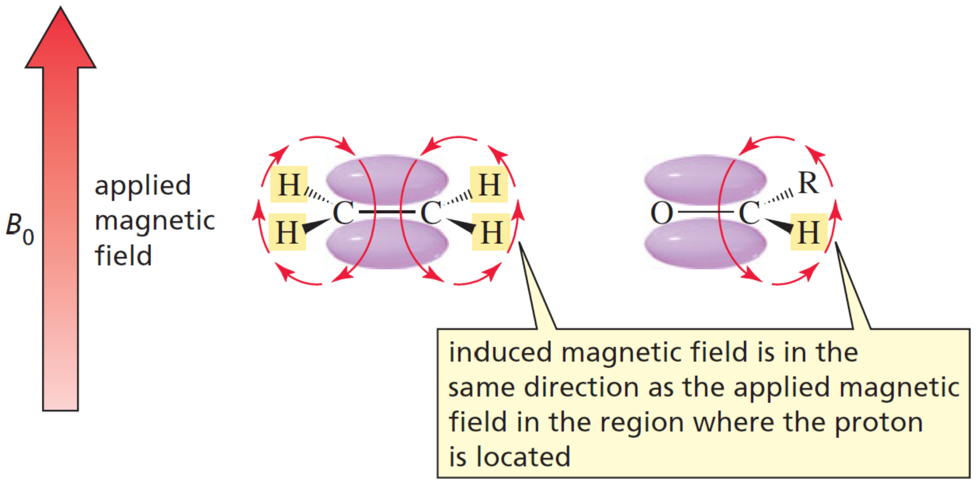


The magnetic field induced by the π electrons of a benzene ring in the region where **benzene's protons are located is oriented in the same direction** as the applied magnetic field. As a result, the protons sense **a larger effective magnetic field**-namely, the sum of the strengths of the applied field and the induced field. Because frequency is proportional to the strength of the magnetic field experienced by the protons, **the protons show signals at *higher frequencies*** than they would if the p electrons did not induce a magnetic field.

NMR Spectroscopy

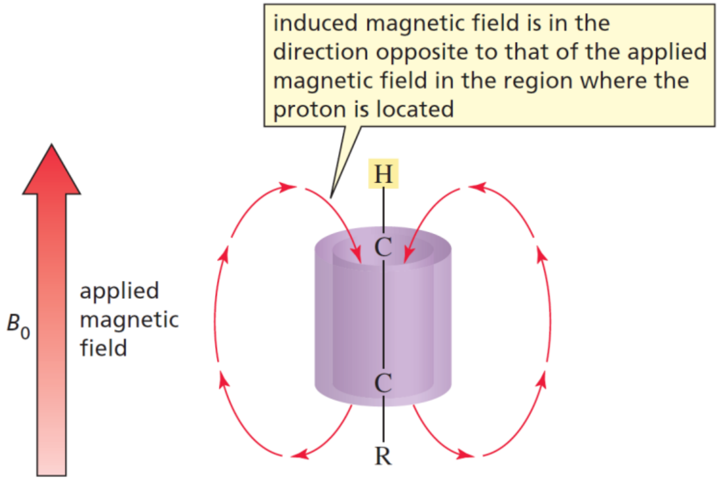
DIAMAGNETIC ANISOTROPY

The magnetic field induced by the p electrons of an alkene or by the π electrons of an aldehyde (in the region where protons bonded to sp^2 carbons are located) is also oriented in the ***same direction* as the applied magnetic field**. These protons, too, show signals at **higher than expected** frequencies.



In contrast, the chemical shift of a hydrogen bonded to an *sp* carbon is at a **lower frequency** (~1.9 ppm) than one would predict from the electronegativity of the *sp* carbon.

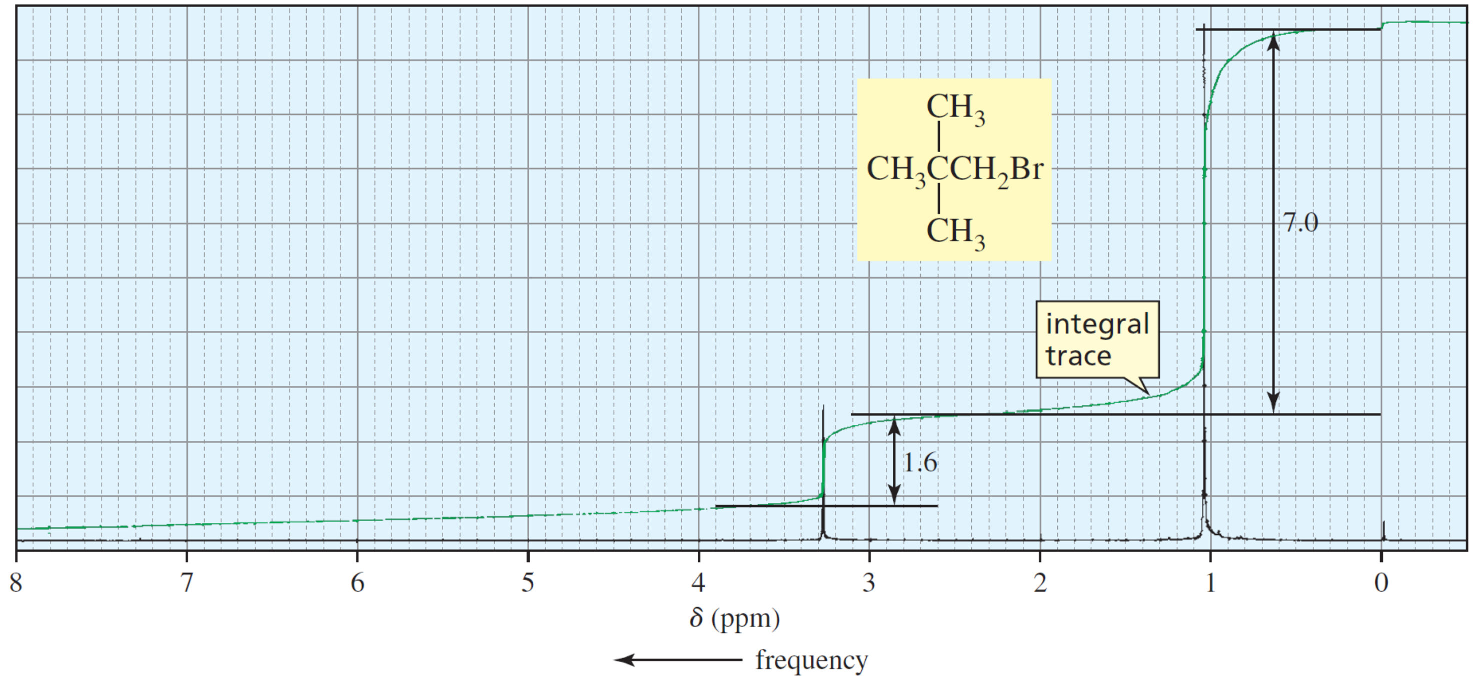
This is because the direction of the magnetic field induced by the alkyne's cylinder of π electrons, in the region where the proton is located, **is *opposite* to the direction of the applied magnetic field**. Thus, the proton senses a smaller effective magnetic field and therefore shows a signal at a *lower frequency* than it would if the π electrons did not induce a magnetic field.



NMR Spectroscopy

THE INTEGRATION OF NMR SIGNALS REVEALS THE RELATIVE NUMBER OF PROTONS CAUSING EACH SIGNAL

- The two signals in the ^{1}H NMR spectrum in are not the same size because ***the area under each signal is proportional to the number of protons producing the signal*** .
- The area under the signal occurring at the lower frequency is larger because the signal is produced by *nine* methyl protons, whereas the smaller, higher-frequency signal is produced by *two* methylene protons.



Figure

Analysis of the integration line in the ^{1}H NMR spectrum of 1-bromo-2,2-dimethylpropane. The peak at 3.3 ppm has a smaller integral trace than the peak at 1.0 ppm because the peak at 3.3 ppm is produced by two methylene protons, whereas the peak at 1.0 ppm is produced by nine methyl protons.

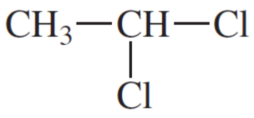
NMR Spectroscopy

THE INTEGRATION OF NMR SIGNALS REVEALS THE RELATIVE NUMBER OF PROTONS CAUSING EACH SIGNAL

An NMR spectrometer is equipped with a computer that calculates the integrals electronically and then displays them as an integral trace superimposed on the original spectrum. The **height of each step in the integral trace is proportional to the area under the corresponding signal, which, in turn, is proportional to the number of protons** producing the signal.

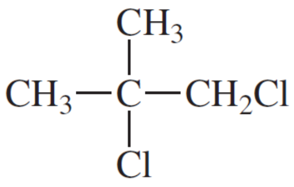
For example, the heights of the integration steps in the previous figure tell us that the ratio of the integrals is approximately 1.6 : 7.0. Dividing by the smallest number gives a new ratio (1 : 4.4). We **then need to multiply this ratio by a number that will make all the numbers in the ratio close to whole numbers**—in this case, we multiply by 2. This means that the ratio of protons in the compound is 2 : 8.8, which is rounded to 2 : 9, **since there can be only whole numbers of protons**. (The measured integrals are approximate because of experimental error.)

Integration tells us the ***relative* number of protons** that produce each signal, not the *absolute* number. In other words, integration could not distinguish between the following two compounds because both would show an integral ratio of 1 : 3.



1,1-dichloroethane

ratio of protons = 1 : 3



1,2-dichloro-2-methylpropane

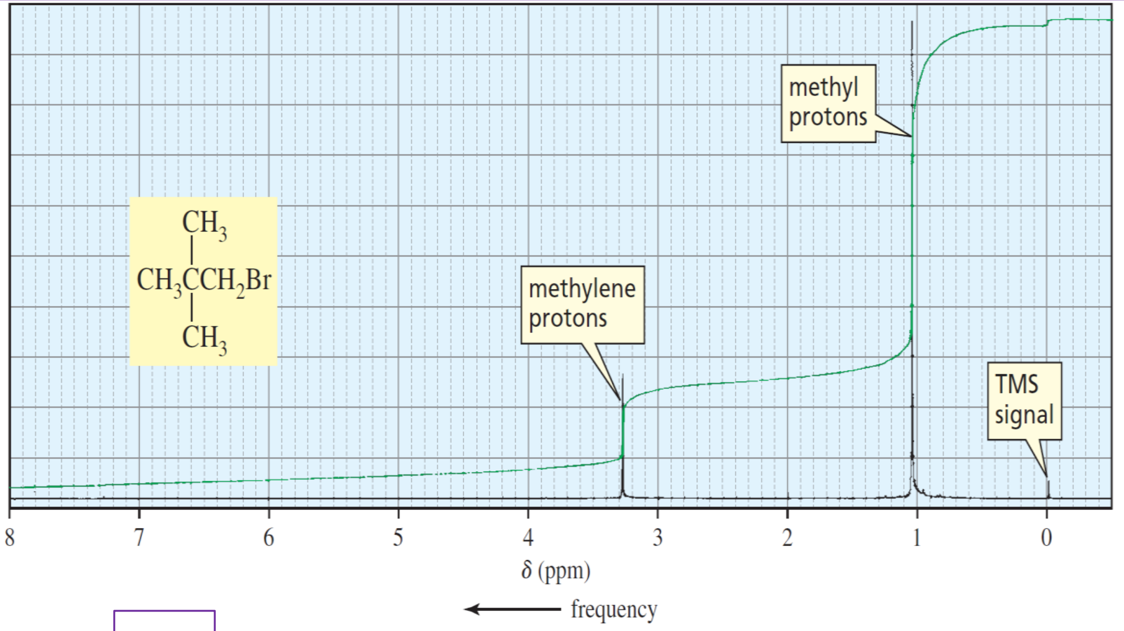
ratio of protons = 2 : 6 = 1 : 3

NMR Spectroscopy

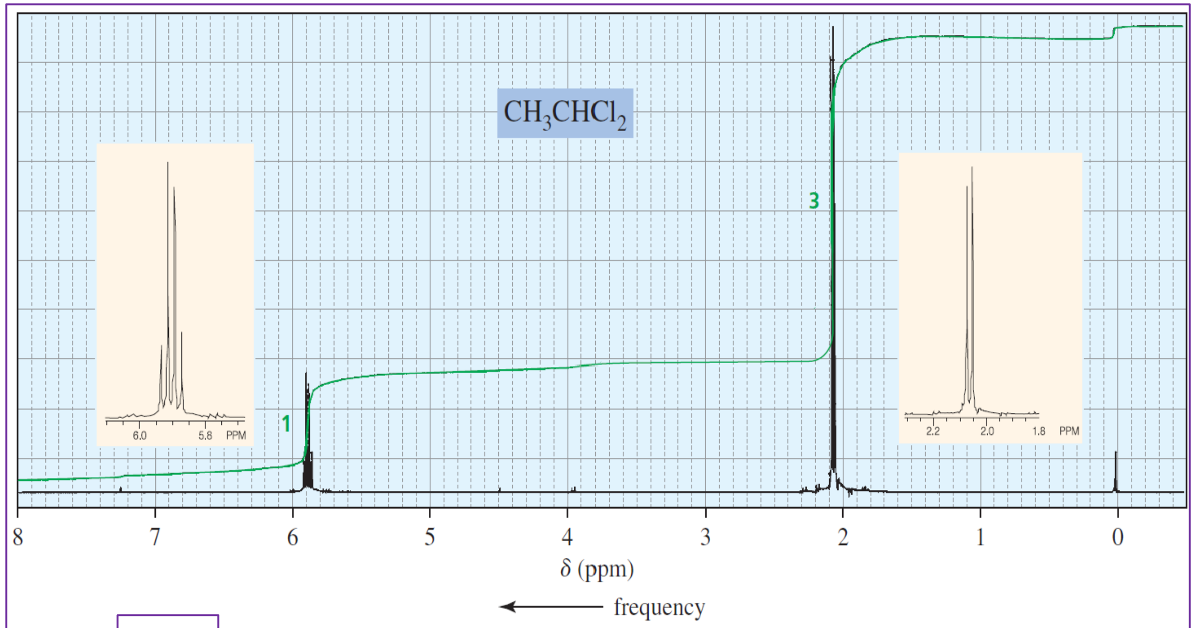
THE SPLITTING OF SIGNALS IS DESCRIBED BY THE *N* + 1 RULE

Notice that the shapes of the signals in the ^{1}H NMR spectrum in **Figure B** are **different** from the shapes of the signals in the ^{1}H NMR spectrum in **Figure A** . Both signals in **Figure A** are **singlets,** meaning each is composed of a single peak. In contrast, the signal for the methyl protons in **Figure B** (the lower-frequency signal) is split into two peaks (a **doublet**), and the signal for the methine proton is split into four peaks (a **quartet**).

Splitting is caused by ***protons bonded to adjacent carbons***. The splitting of a signal is described by the **N + 1 rule**, where **N** is the **number of equivalent protons** bonded to adjacent carbons that are not equivalent to the proton producing the signal.



▲ Figure A



▲ Figure B

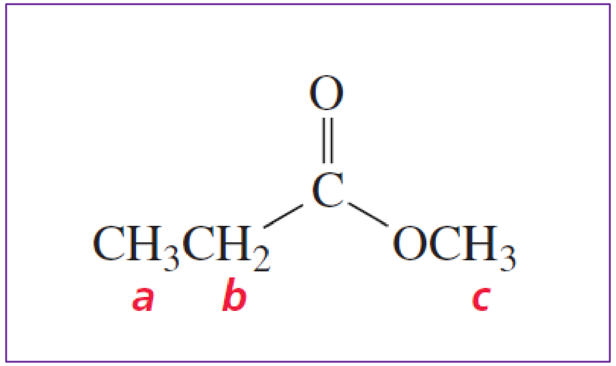
NMR Spectroscopy

THE SPLITTING OF SIGNALS IS DESCRIBED BY THE *N* + 1 RULE

- The number of peaks in a signal is called the **multiplicity** of the signal. **Splitting is always mutual**: if the ***a*** protons split the ***b*** protons, then the ***b*** protons must split the ***a*** protons. The ***a*** and ***b*** protons, in this case, are *coupled protons*. **Coupled protons** split each other's signal. Notice that coupled protons are bonded to adjacent carbons.

- *Keep in mind that it is not the number of protons producing a signal that determines the multiplicity of the signal; rather, it is the* ***number of protons bonded to the immediately adjacent carbons*** *that determines the multiplicity*.

- For example, the signal for the ***a*** protons in the following compound will be split into three peaks (a **triplet**) because the adjacent carbon is bonded to two protons. The signal for the ***b*** protons will appear as a quartet because the adjacent carbon is bonded to three protons, and the signal for the ***c*** protons will be a singlet.



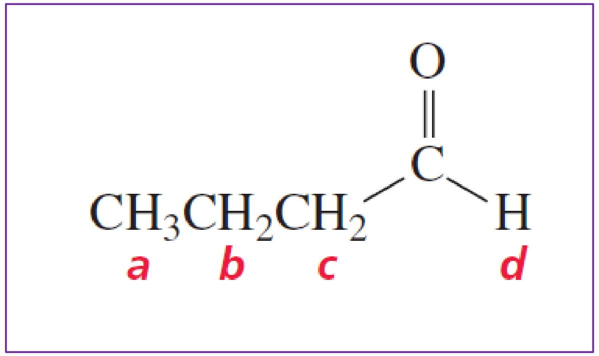
NMR Spectroscopy

THE SPLITTING OF SIGNALS IS DESCRIBED BY THE *N* + 1 RULE

The signal for the ***a*** protons in the following compound will be a **triplet**, the signal for the ***c*** protons will also be a **triplet**, and the signal for the ***d*** proton will be a **singlet** since there are no hydrogens bonded to the adjacent carbon.

Because the ***a*** and ***c*** protons are not equivalent, the $N + 1$ rule has to be applied separately to each set to determine the splitting of the ***b*** protons. Thus, the signal for the ***b*** protons will be **split into a quartet** by the ***a*** protons, and each of the four peaks of **the quartet will be split into a triplet** by the ***c*** protons: $(N_a+1)(N_b+1) = (4)(3) = 12$.

As a result, the signal for the ***b*** protons is a **multiplet** (a signal that is **more complex than a triplet, quartet**, etc.). The actual number of peaks seen in the multiplet depends on how many of them overlap.



NMR Spectroscopy

THE SPLITTING OF SIGNALS IS DESCRIBED BY THE *N* + 1 RULE

A signal for a proton is never split by *equivalent* protons.

For example, the ^{1}H NMR spectrum of bromomethane shows **one singlet**. The three methyl protons are chemically equivalent, and chemically equivalent protons do not split each other's signal. The four protons in 1,2-dichloroethane are also chemically equivalent, so its ^{1}H NMR spectrum also shows one singlet.

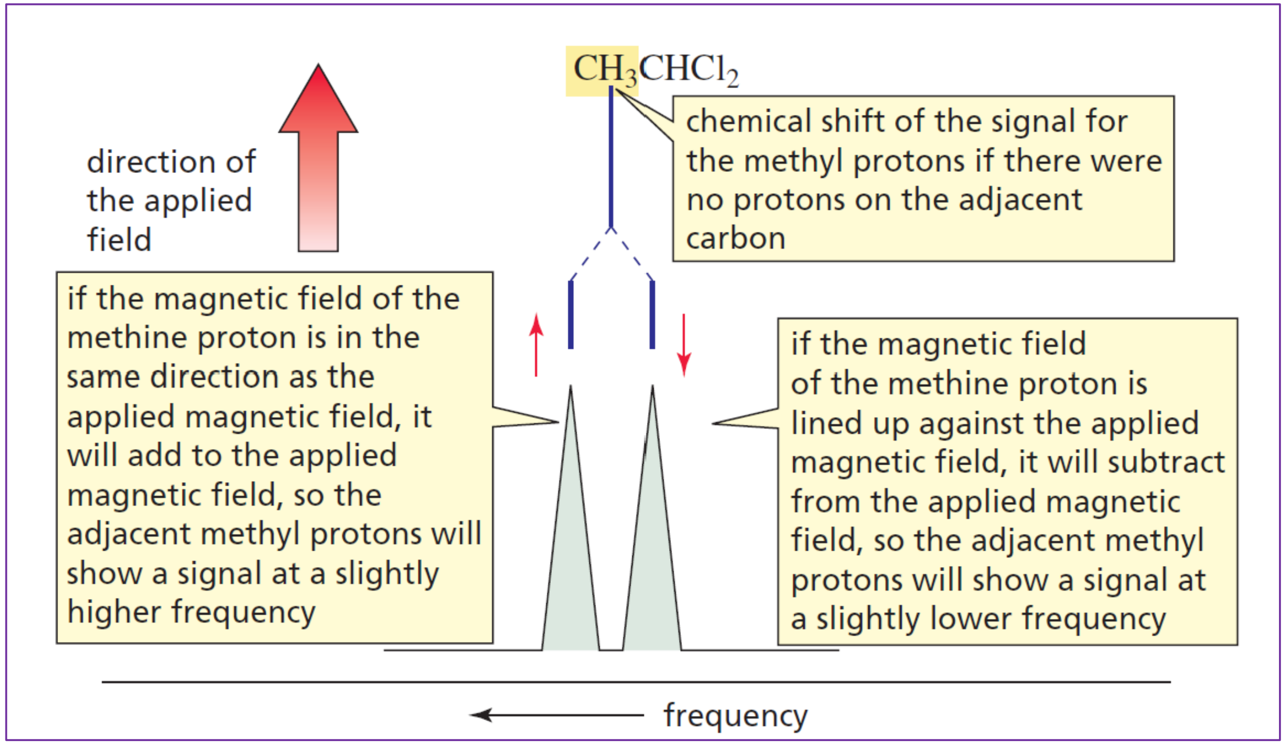
CH_3Br	$ClCH_2CH_2Cl$
bromomethane	**1,2-dichloroethane**

each compound shows one singlet in its ^{1}H NMR spectrum because equivalent protons do not split each other's signals

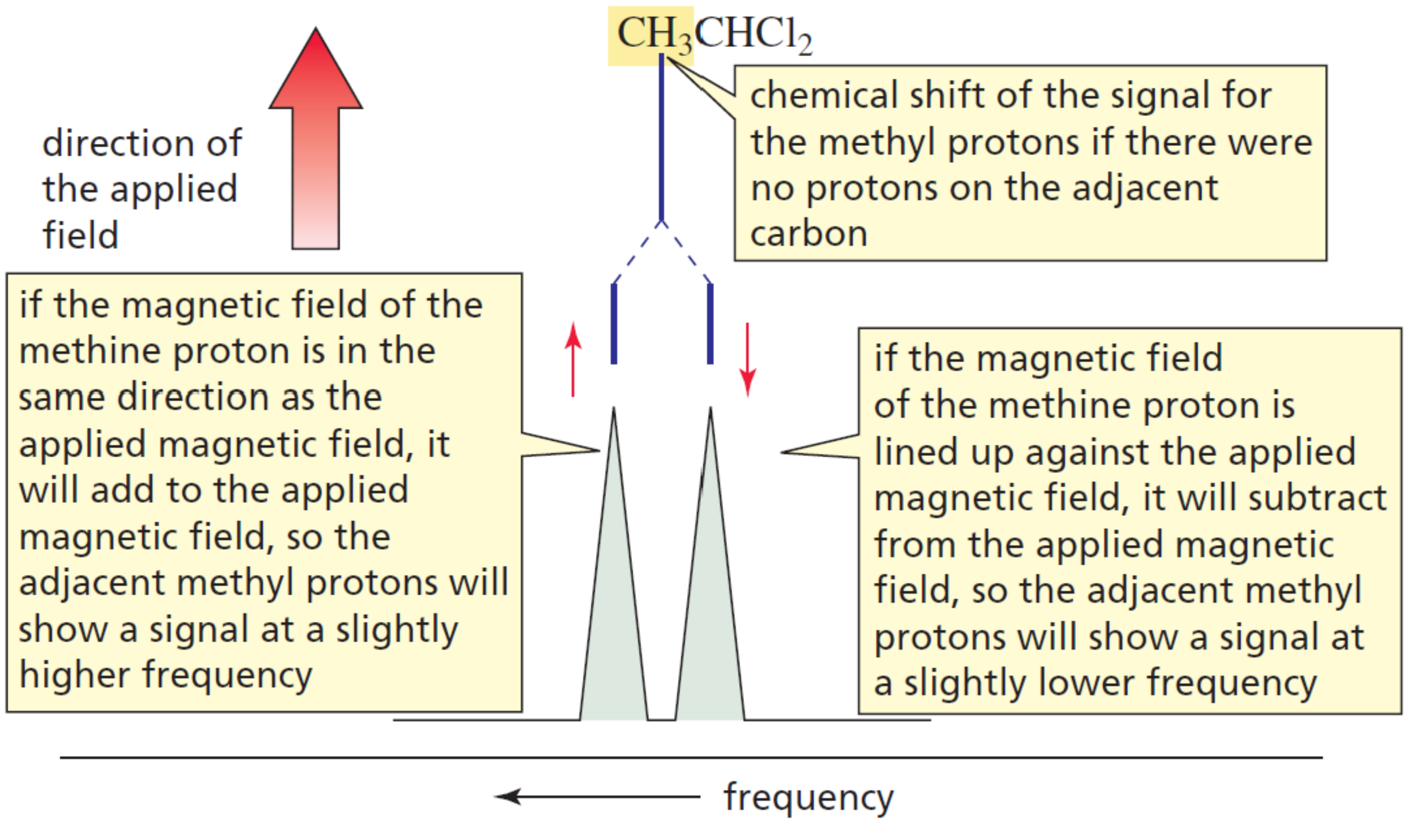
Splitting occurs when different kinds of protons are **close enough for their magnetic fields to influence** one another- a situation called **spin–spin coupling .**

For example, the frequency at which the **methyl protons** of **1,1-dichloroethane** show a signal is influenced by the **magnetic field of the methine proton** (Figure).

- If the magnetic field of the methine proton **aligns** *with* that of the applied magnetic field (*indicated in Figure by the up arrow*), then it will **add to the applied magnetic field**, which will cause the methyl protons to show a **signal at a slightly higher frequency**.
- On the other hand, if the magnetic field of the methine proton **aligns *against*** the applied magnetic field (indicated in Figure by the down arrow), then it will subtract from the applied magnetic field and the methyl protons will show a signal at a lower frequency.



Therefore, the signal for the methyl protons is split into two peaks, one at a higher frequency and one at a lower frequency. Because the **α** - and **β** -spin states have almost the same population, about **half the methine protons** are lined up with the applied magnetic field and about **half are lined up against it**. As a result, the two peaks of the ***doublet*** have approximately the same height and area.



NMR Spectroscopy

WHAT CAUSES SPLITTING?

Similarly, the frequency at which the **methine proton shows a signal** is influenced by the magnetic fields of the **three protons bonded to the adjacent carbon**. The magnetic fields of all three methyl protons can align with the applied magnetic field, two can align with the field and one against it, one can align with it and two against it, or all three can align against it (Figure A). Because the magnetic field that the methine proton senses is affected in four different ways, its signal is a *quartet* (Figure B).

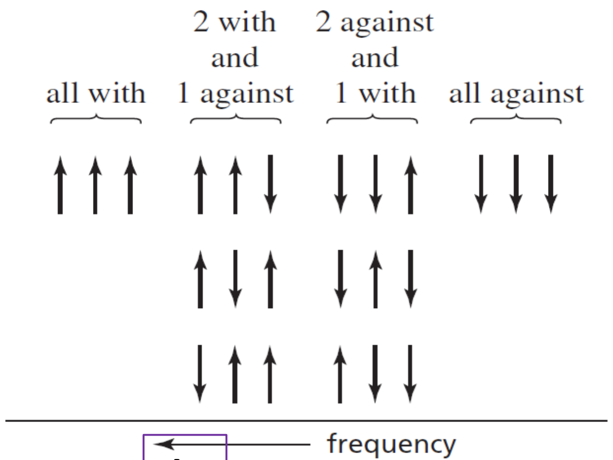


Figure A
The different ways in which the magnetic fields of three protons can be aligned.

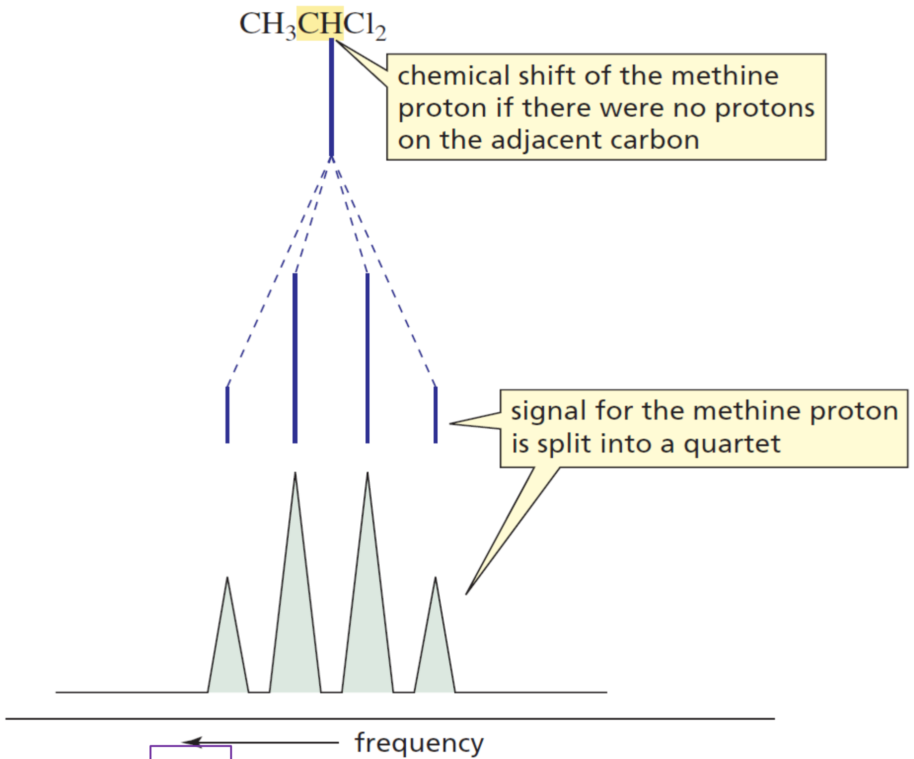


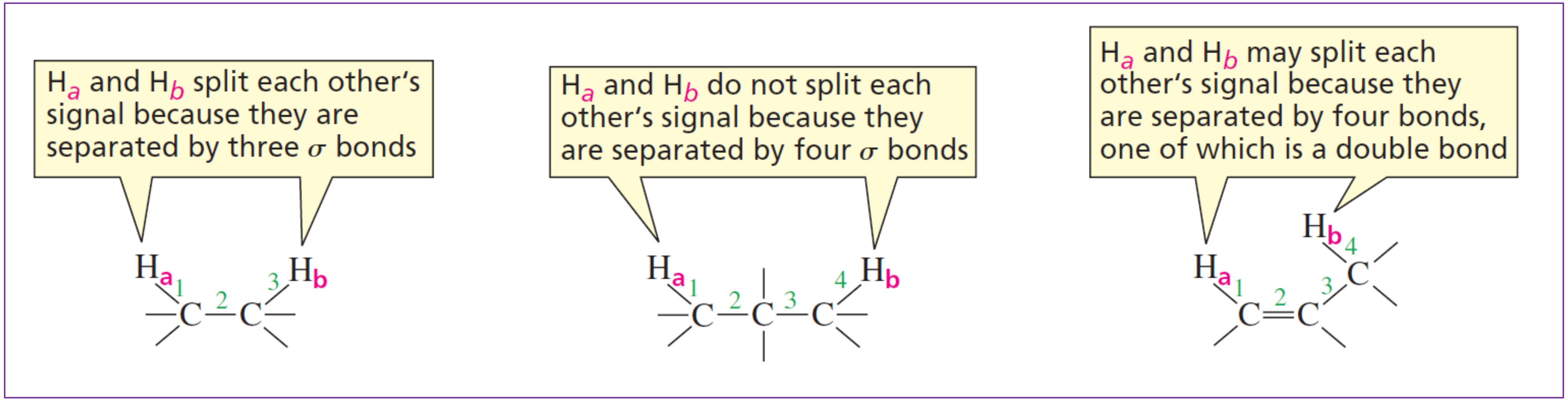
Figure B
The signal for the methine proton of 1,1-dichloroethane is split into a quartet by the methyl protons.

NMR Spectroscopy

WHAT CAUSES SPLITTING?

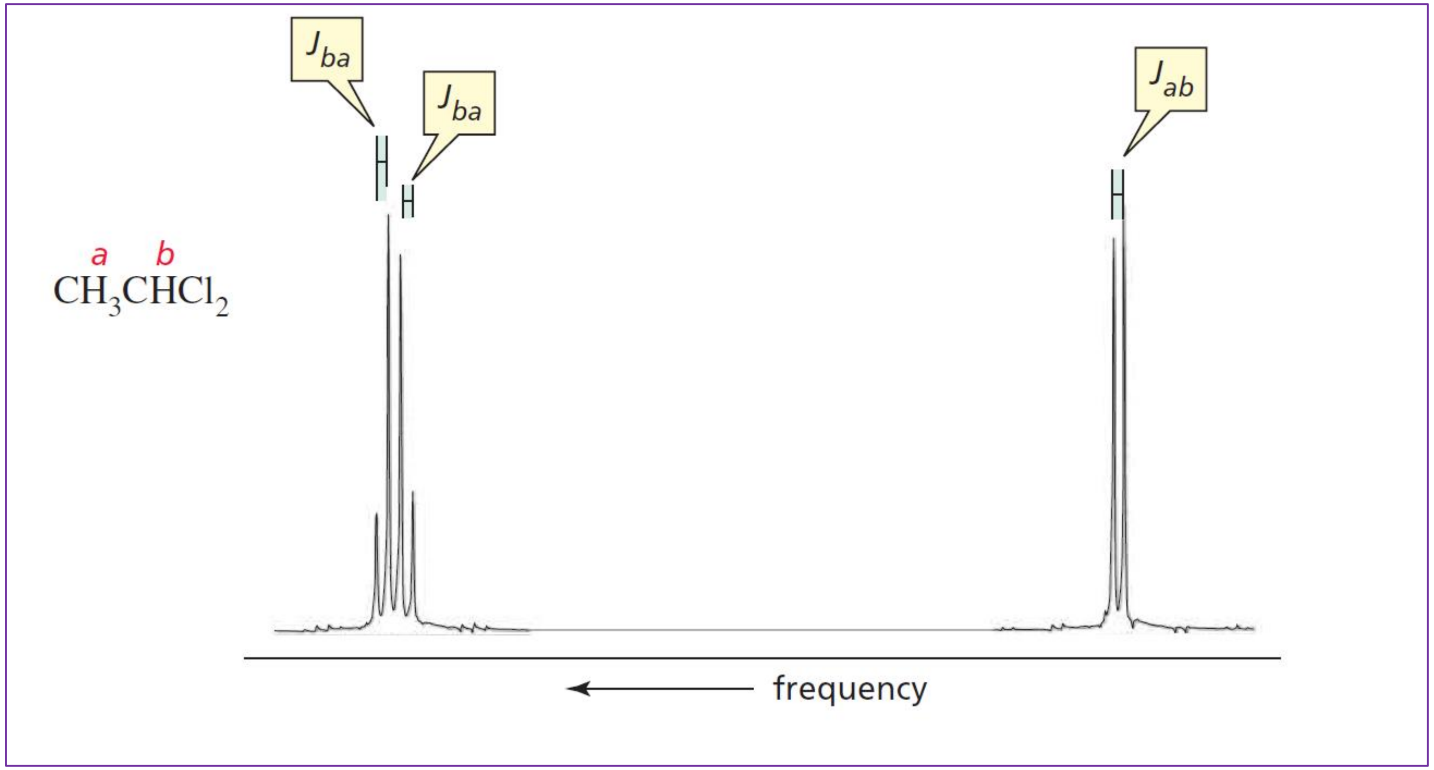
Normally, *nonequivalent* protons split each other's signal only if they are on the same or *adjacent* carbons. Splitting is a "***through-bond***" effect, not a "***through-space***" effect, and it is rarely observed if the protons are separated by more than three **σ** bonds.

However, if they are separated by more than three bonds and one of the bonds is a double or a triple bond, then splitting is sometimes observed. This phenomenon is called **long-range coupling .**



COUPLING CONSTANTS IDENTIFY COUPLED PROTONS

- The distance, in hertz, between two adjacent peaks of a split NMR signal is called the **coupling constant** (denoted by ***J***).
- The coupling constant for the ***a*** protons being split by the ***b*** proton is denoted by $\boldsymbol{J_{ab}}$.
- The signals of coupled protons (protons that split each other's signal) have the same coupling constant; in other words, $\boldsymbol{J_{ab} = J_{ba}}$.
- Coupling constants are useful in analyzing complex NMR spectra because protons on adjacent carbons can be identified by their identical coupling constants.

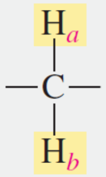
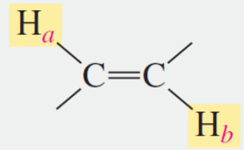
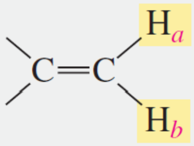
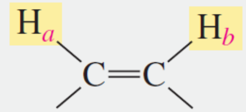
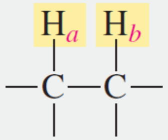
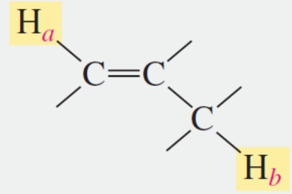
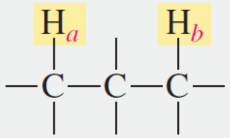


The ***a*** and ***b*** protons of 1,1-dichloroethane are coupled protons—that is, they split each others signals, so their signals have the same coupling constant, $J_{ab} = J_{ba}$.

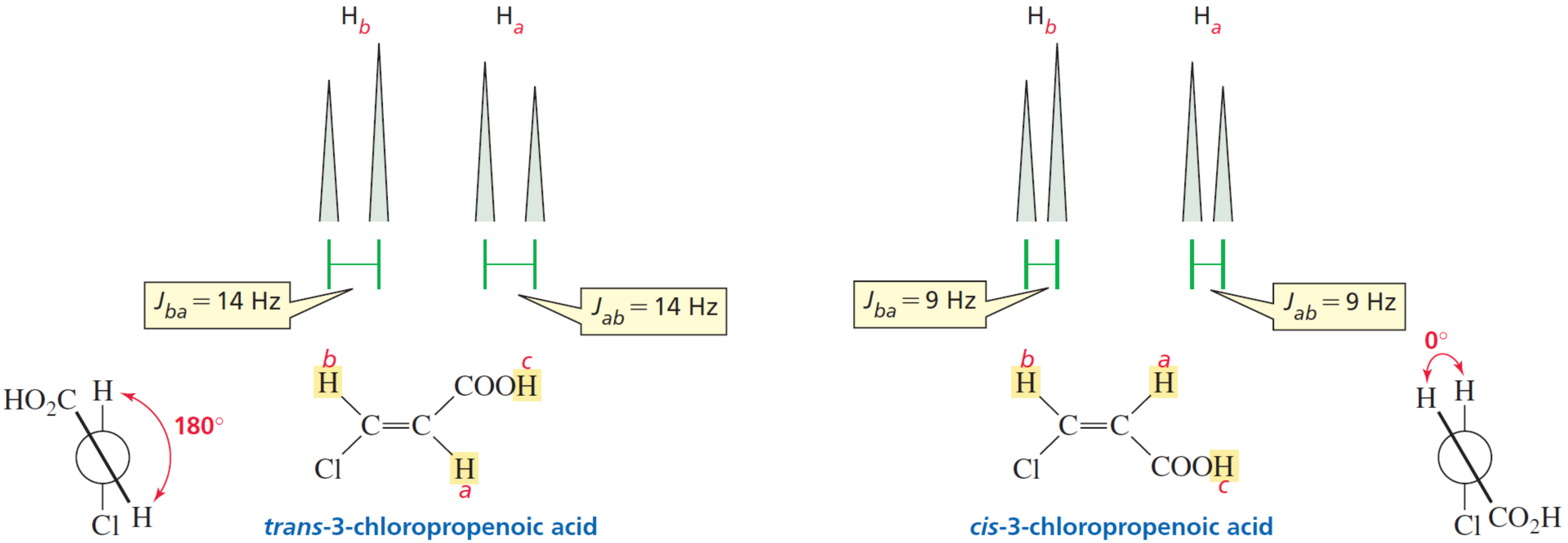
NMR Spectroscopy

- *J* has the **same value regardless of the operating frequency** of the spectrometer.
- The **magnitude of a coupling constant is a measure of how strongly the nuclear spins** of the coupled protons influence each other.
- It depends, therefore, on **the number of bonds and the type of bonds** that connect the coupled protons, as well as on the geometric relationship of the protons.

Characteristic coupling constants are shown in Table Below; they range from **0 to 15 Hz**.

Types of protons	Approximate value of J_{ab} (Hz)	Types of protons	Approximate value of J_{ab} (Hz)
	12		15 (trans)
	2		10 (cis)
	7		1 (long-range coupling)
	0		

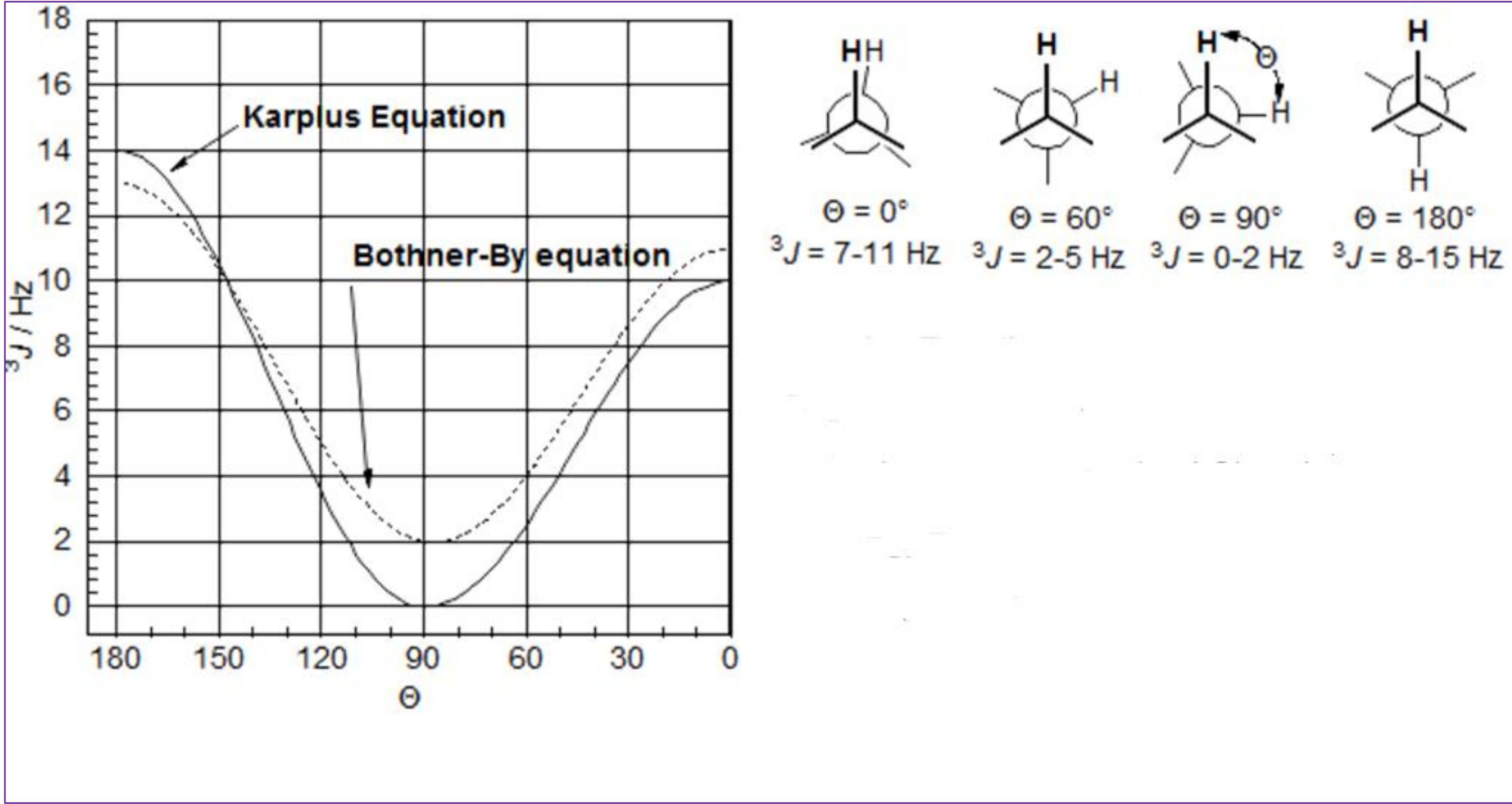
Coupling constants can be used to distinguish between the ^{1}H NMR spectra of ***cis*** and ***trans*** alkenes. The coupling constant of ***trans*** **vinylic protons is significantly greater than that of *cis* vinylic protons**, because the coupling constant depends on the dihedral angle between the two **C-H** bonds in the **H-C=C-H** unit. The coupling constant is greatest when the angle between the two C-H bonds is **180 (*trans*)** and smallest when the angle is **0 (*cis*).**



▲ **Figure**

The doublets observed for the ***a*** and ***b*** protons in the ^{1}H NMR spectra of a trans alkene and a cis alkene. The coupling constant for trans protons (14 Hz) is greater than the coupling constant for cis protons (9 Hz) because it depends on the dihedral angle (180° for trans protons; 0° for cis protons).

NMR Spectroscopy

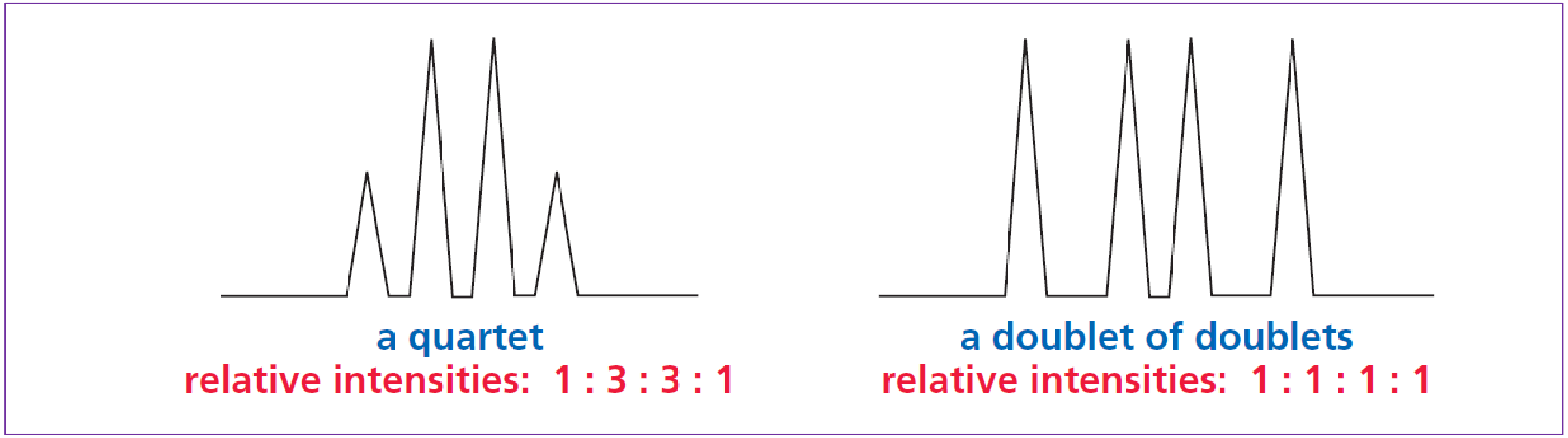


Overlap is maximized when the two, vicinal C–H bonds are aligned in an eclipsed conformation (H–C–C–H dihedral angle =0°) or anti (H–C–C–H dihedral angle =180°) conformation. When the dihedral angle is ~90°the overlap of the two vicinal C–H bonds is minimized.

NMR Spectroscopy

There is a clear difference between a **quartet** and a **doublet of doublets**, even though both have four peaks. A quartet results from splitting by *three equivalent* adjacent protons; it therefore has relative peak intensities of 1 : 3 : 3 : 1, and the individual peaks are equally spaced.

A doublet of doublets, on the other hand, results from splitting by *two nonequivalent* adjacent protons; it has relative peak intensities of 1 : 1 : 1 : 1, and the individual peaks are not necessarily equally spaced



NMR Spectroscopy

Let's now summarize the kind of information that can be obtained from an ^1H NMR spectrum:

1. The number of signals indicates the number of different kinds of protons in the compound.
2. The position of a signal indicates the kind of proton(s) that produce the signal (methyl, methylene, methine, allylic, vinylic, benzene, and so on) and the kinds of neighboring substituents.
3. The integration of the signal tells the relative number of protons that produce the signal.
4. The multiplicity of the signal ($N + 1$) tells the number of protons (N) bonded to adjacent carbons.
5. The coupling constants identify coupled protons.